ExpirySense

SSL Certificate Expiry Watcher

USE CASE DOCUMENT

Document Type	Project
Use Case Specification	ExpirySense

1. Introduction

1.1 Project Overview

ExpirySense is a real-time SSL/TLS certificate monitoring web application that enables users to track certificate expiry status across multiple domains. It provides instant visual dashboards, risk classification, AI-powered remediation recommendations via the Groq LLM (llama3-8b-8192), and exportable reports.

1.2 Purpose of This Document

This document defines all functional use cases of the ExpirySense system, describing how various actors interact with the system to accomplish specific goals. It serves as the primary reference for developers, testers, and stakeholders.

1.3 Scope

In Scope:

- SSL/TLS certificate scanning for user-submitted domains
- Real-time expiry classification (Critical / Warning / Healthy)
- Dashboard analytics and risk distribution visualization
- CSV bulk upload and report export
- AI-powered remediation workbook generation
- SQLite-backed persistent scan log
- Individual hostname management (add / delete)

Out of Scope:

- Automated certificate renewal or provisioning
- User authentication and multi-tenancy
- Email/SMS alert notifications
- Third-party CA integrations

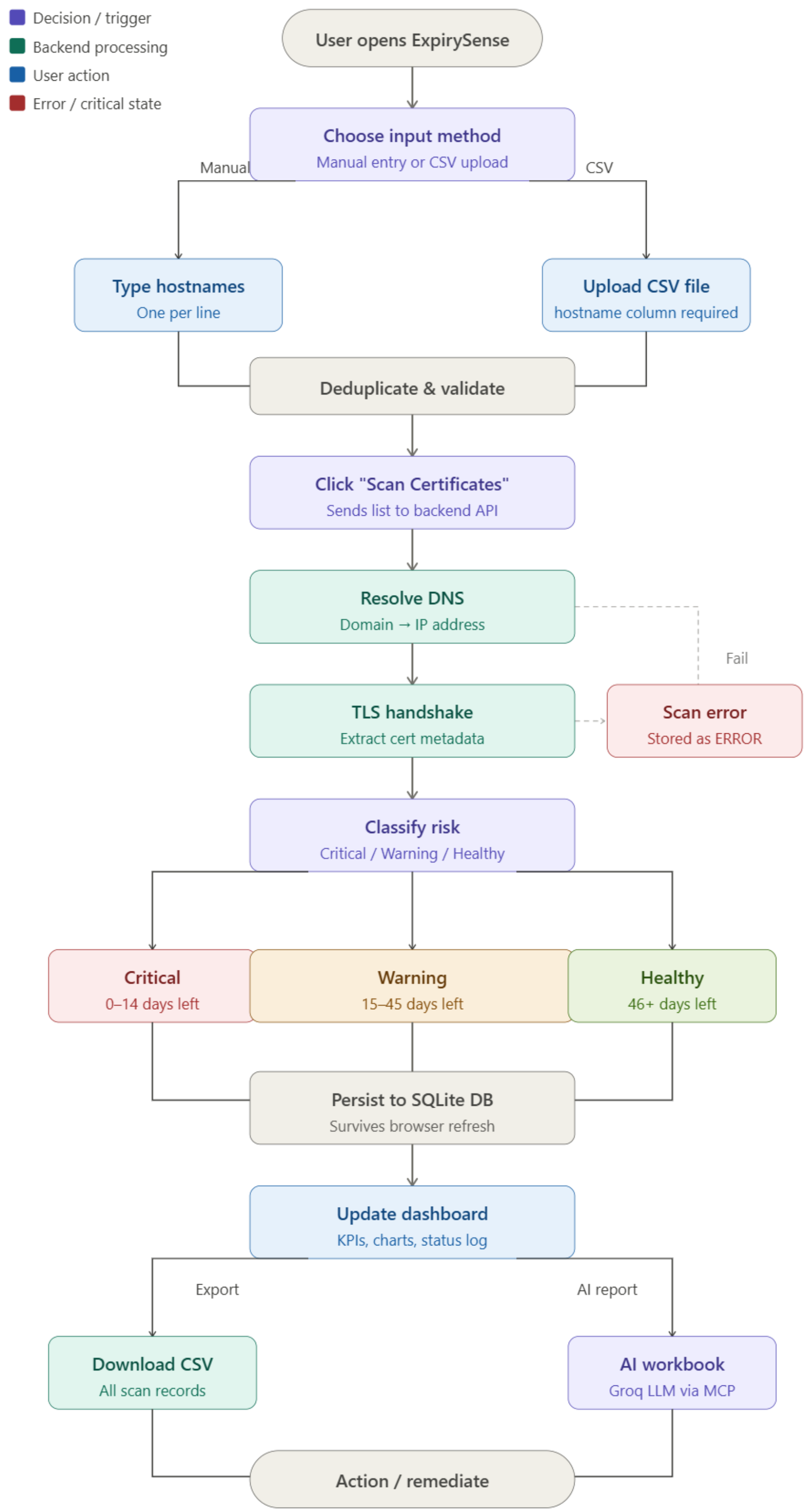
2. Actors

The following actors interact with the ExpirySense system:

Actor ID	Actor Name	Description
ACT-01	**End User**	A system administrator, DevOps engineer, or security analyst who interacts with the web UI to monitor SSL certificates.
ACT-02	**ExpirySense Backend**	The Python/FastAPI server that processes scan requests, connects to target hosts, and persists results in SQLite.
ACT-03	**Target Host**	Any external domain or IP whose SSL certificate is being probed (e.g., google.com, github.com).
ACT-04	**Groq LLM API**	The external AI inference system (llama3-8b-8192) that generates incident analysis and remediation guidance.
ACT-05	**MCP Server Layer**	The Model Context Protocol layer that brokers structured tool calls between the backend and the AI model.
ACT-06	**SQLite Database**	Persistent storage that maintains all scan records across browser sessions.

3. Use Case Summary

ID	Use Case Name	Primary Actor(s)	Priority
UC-01	Add Hostname Manually	End User	High
UC-02	Upload CSV Hostnames	End User	High
UC-03	Scan SSL Certificates	End User / Backend	Critical
UC-04	View Dashboard Analytics	End User	High
UC-05	View Certificate Status Log	End User	Medium
UC-06	Download CSV Report	End User	Medium
UC-07	Download AI Remediation Workbook	End User / Groq LLM	High
UC-08	Delete Hostname	End User	Medium
UC-09	View Risk Distribution Chart	End User	Low
UC-10	View Expiry Timeline	End User	Low
UC-11	Monitor Backend Connectivity	System	High


Decision / trigger
Backend processing
User action
Error / critical state
User opens ExpirySense
Choose input method
Manual entry or CSV upload
Manual
CSV
Type hostnames
One per line
Upload CSV file
hostname column required
Deduplicate & validate
Click "Scan Certificates"
Sends list to backend API
Resolve DNS
Domain → IP address
Fail
TLS handshake
Extract cert metadata
Scan error
Stored as ERROR
Classify risk
Critical / Warning / Healthy
Critical
0–14 days left
Warning
15–45 days left
Healthy
46+ days left
Persist to SQLite DB
Survives browser refresh
Update dashboard
KPIs, charts, status log
Export
AI report
Download CSV
All scan records
AI workbook
Groq LLM via MCP
Action / remediate

4. Detailed Use Cases

UC-01 — Add Hostname Manually

Use Case ID	**UC-01**
Use Case Name	**Add Hostname Manually**
Description	The End User types one or more domain names into the manual entry text area to queue them for SSL scanning.
Primary Actors	End User
Preconditions	The ExpirySense web interface is open and backend is reachable.
Postconditions	Entered hostnames are staged in the input queue, ready for scanning.

Main Success Flow:

1	User navigates to the Target Hostnames section.
2	User selects the Manual Entry tab.
3	User types one or more hostnames, one per line (e.g., google.com, github.com).
4	System automatically excludes duplicates and blank lines.
5	User clicks Scan Certificates to initiate the scan.

Alternative Flows:

- A1: User switches to CSV Upload tab instead — proceeds to UC-02.
- A2: User clicks Reset to clear all staged hostnames before scanning.

Exception Flows:

- E1: User enters an invalid hostname format — backend returns a connection error; status shown as ERROR in the log.
- E2: All entered hostnames are duplicates — system stages zero unique entries; scan produces no new records.

UC-02 — Upload CSV Hostnames

Use Case ID	**UC-02**
Use Case Name	**Upload CSV Hostnames**

Description	The End User uploads a CSV file containing a list of hostnames for bulk SSL certificate scanning.
Primary Actors	End User
Preconditions	User has a CSV file with a hostname column header and at least one domain entry.
Postconditions	All valid hostnames from the CSV are staged for scanning.

Main Success Flow:

1	User selects the CSV Upload tab in the Target Hostnames section.
2	User drags and drops a CSV file onto the upload zone, or clicks to browse.
3	System validates the file format — confirms .csv extension and presence of hostname column.
4	System parses the CSV, strips duplicates and blank values.
5	System displays success confirmation: 'CSV columns read successfully!'
6	User clicks Scan Certificates to proceed.

Alternative Flows:

- A1: User re-uploads a different CSV file — system replaces the previously staged list.
- A2: User switches back to Manual Entry tab — CSV data is discarded.

Exception Flows:

- E1: File is not a .csv format — system rejects the upload and shows an error.
- E2: CSV does not contain a hostname column — system alerts the user with the expected format example.
- E3: CSV is empty after deduplication — system stages zero hostnames.

UC-03 — Scan SSL Certificates

Use Case ID	**UC-03**
Use Case Name	**Scan SSL Certificates**
Description	The system initiates real-time TLS connections to all staged hostnames, retrieves certificate metadata, classifies expiry risk, and persists results.
Primary Actors	End User, ExpirySense Backend, Target Host, SQLite Database
Preconditions	At least one valid hostname is staged. Backend is online. DNS is reachable.
Postconditions	Scan records are persisted in SQLite. Dashboard counters and status log are updated.

Main Success Flow:

1	User clicks the Scan Certificates button.
2	Frontend sends the hostname list to the ExpirySense backend via API.
3	Backend resolves DNS for each hostname (Resolve Domain Name Services).
4	Backend negotiates a TLS handshake with each target host (Negotiate Cryptographic Handshake).
5	Backend extracts certificate metadata: issuer, expiry date, TLS protocol version.
6	Backend calculates days remaining until expiry and assigns a risk classification: Critical (0–14 days), Warning (15–45 days), or Healthy (46+ days).
7	Results are written to the SQLite database.
8	Backend queries the Groq LLM via MCP for AI incident analysis (async).
9	Frontend polls the backend and renders updated dashboard counters and status log entries.

Alternative Flows:

- A1: Hostname already exists in the database — backend updates the existing record with fresh scan data.
- A2: Groq LLM is unavailable — scan completes without AI analysis; remediation workbook is unavailable.

Exception Flows:

- E1: Backend is offline — frontend shows 'Connection error to ExpirySense backend. Monitoring offline.' banner.
- E2: TLS handshake fails (host unreachable, refused connection) — record is stored with status ERROR.
- E3: Certificate is expired — days remaining shown as 0 or negative; classified as Critical.

UC-04 — View Dashboard Analytics

Use Case ID	**UC-04**
Use Case Name	**View Dashboard Analytics**
Description	The End User views aggregated certificate health metrics including total monitored domains and counts per risk tier.
Primary Actors	End User
Preconditions	At least one scan has been executed and records exist in the database.
Postconditions	User has a clear overview of the SSL health posture across all monitored domains.

Main Success Flow:

1	User opens the ExpirySense dashboard.
2	System fetches current scan records from the SQLite database.
3	Dashboard renders four KPI tiles: Total Hostnames, Critical, Warning, Healthy.
4	Risk Distribution doughnut chart shows proportional breakdown of certificate statuses.
5	Expiry Timeline histogram shows distribution of remaining days across all certificates.
6	User reads the metrics to prioritise renewal actions.

Alternative Flows:

- A1: No scans have been run yet — all counters show 0 and charts display 'No scan data available'.

Exception Flows:

- E1: Backend sync fails — dashboard may display stale data; 'Scan Sync: Connected' indicator turns red.

UC-06 — Download CSV Report

Use Case ID	**UC-06**
Use Case Name	**Download CSV Report**
Description	The End User exports all current scan records as a structured CSV file for offline analysis or audit purposes.
Primary Actors	End User
Preconditions	At least one scan record exists in the database.
Postconditions	A CSV file containing all scan records is downloaded to the user's device.

Main Success Flow:

1	User clicks the Download CSV Report button in the top action bar.
2	Backend compiles all scan records from SQLite into CSV format.
3	System serves the file as a browser download.
4	User saves the CSV file locally.

Alternative Flows:

- A1: User opens the CSV in Excel or a spreadsheet tool for further analysis.

Exception Flows:

- E1: No records exist — downloaded CSV contains only the header row.

UC-07 — Download AI Remediation Workbook

Use Case ID	**UC-07**
Use Case Name	**Download AI Remediation Workbook**
Description	The End User downloads an AI-generated workbook containing incident analysis and remediation steps for certificates requiring attention.
Primary Actors	End User, Groq LLM API, MCP Server Layer
Preconditions	At least one scan has completed. Groq LLM API is reachable.
Postconditions	A workbook file is downloaded containing AI-generated remediation guidance per affected hostname.

Main Success Flow:

1	User clicks Download AI Remediation Workbook.
2	Backend identifies hostnames with Critical or Warning status.
3	MCP Server Layer sends structured prompts to the Groq llama3-8b-8192 model.
4	Groq LLM returns incident analysis and step-by-step remediation for each affected domain.
5	Backend assembles the AI outputs into a downloadable workbook.
6	System serves the workbook as a browser download.

Alternative Flows:

- A1: All certificates are Healthy — workbook is generated with a confirmation note; no remediation steps required.

Exception Flows:

- E1: Groq API is unreachable — button is disabled or returns an error message; user is advised to retry.
- E2: LLM response timeout — backend returns partial workbook with a timeout notice.

UC-08 — Delete Hostname

Use Case ID	**UC-08**
Use Case Name	**Delete Hostname**
Description	The End User permanently removes a specific hostname and its scan record from the database.
Primary Actors	End User, SQLite Database
Preconditions	The hostname exists in the SSL Certificate Status Log.
Postconditions	The hostname record is permanently deleted. Dashboard counters are updated.

Main Success Flow:

1	User locates the hostname in the SSL Certificate Status Log.
2	User clicks the Delete action for that hostname.
3	System displays a confirmation modal: 'Are you absolutely sure you want to remove [hostname] from the database? This action is irreversible.'
4	User clicks Delete Forever.
5	Backend deletes the record from SQLite.
6	Frontend refreshes the log and dashboard counters.

Alternative Flows:

- A1: User clicks Keep Domain in the confirmation modal — deletion is cancelled; record is preserved.

Exception Flows:

- E1: Backend is offline during deletion — system shows an error; record remains in the database.

5. Use Case Diagram

The diagram below illustrates the primary actors and their relationships to each use case within the ExpirySense system boundary.

End User
(Primary Actor)

ExpirySense System Boundary

UC-01 Add Hostname Manually
UC-02 Upload CSV Hostnames
UC-03 Scan SSL Certificates
UC-04 View Dashboard Analytics
UC-05 View Certificate Status Log
UC-06 Download CSV Report
UC-07 Download AI Remediation Workbook
UC-08 Delete Hostname

Backend

Groq LLM

SQLite DB

Legend:

- Solid left border (blue) = use case within system boundary
- End User initiates all use cases marked UC-01 through UC-08
- UC-03 and UC-07 involve secondary actors: Backend, Groq LLM, Target Host, and SQLite

6. Non-Functional Requirements

ID	Category	Requirement
NFR-01	**Performance**	Scan results for a single hostname must be returned within 10 seconds under normal network conditions.
NFR-02	**Availability**	The application backend should maintain 99% uptime; frontend shows connectivity status in real time.
NFR-03	**Scalability**	Bulk CSV uploads should support at least 500 hostnames per scan batch without timeout.
NFR-04	**Data Persistence**	All scan records must survive browser refresh and session termination via SQLite persistence.
NFR-05	**Security**	All TLS handshakes are read-only probes; no certificate modification or CA interaction is performed.
NFR-06	**Usability**	The UI must clearly differentiate Critical / Warning / Healthy states through distinct colour coding.
NFR-07	**AI Reliability**	If the Groq LLM API is unavailable, core scanning functions must remain fully operational.

7. Assumptions & Constraints

7.1 Assumptions

- Target hostnames are publicly accessible over the internet on port 443.
- The End User has a modern web browser with JavaScript enabled.
- The Groq API key is correctly configured in the backend environment.
- The backend server has unrestricted outbound HTTPS access to scan target hosts.
- CSV files uploaded by users are trusted input; no malware scanning is performed on uploads.

7.2 Constraints

- The application does not support certificate scanning on non-standard TLS ports without manual configuration.
- AI remediation workbooks depend on Groq API rate limits and availability.
- SQLite is a single-file database; concurrent multi-user write scenarios are not supported in v1.0.
- The system is deployed on Render free tier; cold starts may add latency after periods of inactivity.

Glossary

Term	Definition
SSL / TLS	Secure Sockets Layer / Transport Layer Security — cryptographic protocols for secure communication.
Certificate	A digital document issued by a Certificate Authority (CA) verifying domain ownership and enabling HTTPS.
Expiry Date	The date after which an SSL certificate is no longer valid and browsers will display security warnings.
Critical	Certificates expiring within 0–14 days; require immediate renewal action.
Warning	Certificates expiring within 15–45 days; renewal should be scheduled promptly.
Healthy	Certificates with 46 or more days remaining; no immediate action required.
MCP	Model Context Protocol — a structured interface for tool-augmented LLM interactions.
Groq LLM	A high-speed AI inference platform; ExpirySense uses the llama3-8b-8192 model for analysis.
SQLite	A lightweight, serverless relational database used for persisting scan records.
CSV	Comma-Separated Values — a plain-text file format for tabular data.